

INSTALLATION SETUP MANUAL - PICKLEBALL BOOKING SYSTEM

INTEGRATION WITH INTERNET OF THINGS (IOT)

SOFTWARE REQUIREMENTS

Install the following software before running the system:

- Node.js v18 or later - <https://nodejs.org>
- XAMPP (Apache + MySQL) - <https://www.apachefriends.org>
- Visual Studio Code - <https://code.visualstudio.com>
- Arduino IDE v2 - <https://www.arduino.cc/en/software>
- Google Chrome (recommended browser)

DATABASE SETUP

1. Open XAMPP Control Panel and start MySQL.
2. Open phpMyAdmin in your browser: <http://localhost/phpmyadmin>
3. Click the SQL tab and paste the contents of `server/database-setup.sql`, then click Go.
 - This creates the `pickleball\_db` database and all required tables.
4. Once the tables are created, click on pickleball_db in the left panel, go to the SQL tab again, paste the contents of `server/demo-data.sql`, and click Go.
 - This imports demo user accounts and sample bookings.
 - Demo credentials loaded after this step:

Role	Email	Password
Admin	admin@pickleball.com	AdminPass123
User	user@pickleball.com	UserPass123

BACKEND SETUP

1. Open the project folder in Visual Studio Code.

2. Open the integrated terminal and navigate to `server/` folder:

```
```bash  

cd server

```
```

3. Install dependencies:

```
```bash  

npm install

```
```

4. Copy the environment template and configure it:

```
```bash  

cp .env.example .env

```
```

5. Open `server/.env` and fill in your values:

```
```env  

PORT=5000

DB_HOST=localhost

DB_USER=root

DB_PASSWORD=your_mysql_password

DB_NAME=pickleball_db

FRONTEND_URL=http://localhost:5173

NODE_ENV=development

JWT_SECRET=your_secret_key

JWT_EXPIRATION=7d

```
```

6. To generate a secure `JWT\_SECRET`, run:

```
```bash
```

```
node -e "console.log(require('crypto').randomBytes(32).toString('hex'))"
```

```
```
```

7. Start the backend server:

```
```bash
```

```
npm start
```

```
```
```

8. The server will start on <http://localhost:5000>. You should see:

- MySQL Database connected successfully
- Server running at <http://localhost:5000>

FRONTEND SETUP

1. Install dependencies from the root project folder (not `server/`):

```
```bash
```

```
npm install
```

```
```
```

2. Start the frontend development server:

```
```bash
```

```
npm run dev
```

```
```
```

3. Open your browser and access the application: <http://localhost:5173>

IOT INTEGRATION SETUP

1. Install the following libraries in Arduino IDE via Sketch → Include Library → Manage

Libraries:

- ArduinoJson

- Keypad
 - ESP32Servo
2. Install the ESP32 board package via File → Preferences → Additional Board Manager URLs:

https://raw.githubusercontent.com/espressif/arduino-esp32/gh-pages/package_esp32_index.json
 3. Then go to Tools → Board → Boards Manager, search for `ESP32 by Espressif Systems`, and install it.
 4. Open `firmware/pickleball\_gate\_controller/pickleball\_gate\_controller.ino` in Arduino IDE.
 5. Update the Wi-Fi credentials at the top of the file:


```
```cpp  

const char* WIFI_SSID = "YOUR_WIFI_NAME";

const char* WIFI_PASS = "YOUR_WIFI_PASSWORD";
```
  6. Set the backend API URL to your PC's local IP address (find it with `ipconfig` on Windows):  
  

```
```cpp  
  
const char* API_BASE_URL = "http://YOUR_PC_IP:5000";
```
 7. The ESP32 and your PC must be on the same Wi-Fi network.
 8. Set the court number this ESP32 controls (1 or 2):


```
```cpp  

const int COURT_NUMBER = 1;

```
```
 9. In Arduino IDE, select:
 - Board: ESP32 Dev Module

- Port: the correct COM port for your ESP32

10. Click Upload

HARDWARE PIN CONNECTIONS

1. Servo Motor (Gate)

| Servo Wire | ESP32 Pin |
|------------|-----------|
| VCC (Red) | 5V |
| GND | GND |
| Signal | GPIO 18 |

2. PIR Motion Sensor

| PIR Wire | ESP32 Pin |
|----------|-----------|
| VCC | 5V |
| GND | GND |
| OUT | GPIO 4 |

3. 3×4 Matrix Keypad

| Keypad Pin | ESP32 GPIO |
|------------|------------|
| Row 1 | GPIO 13 |
| Row 2 | GPIO 12 |
| Row 3 | GPIO 14 |
| Row 4 | GPIO 27 |
| Column 1 | GPIO 26 |
| Column 2 | GPIO 25 |
| Column 3 | GPIO 33 |

IOT SYSTEM VERIFICATION

1. Open Serial Monitor in Arduino IDE (baud rate: 115200).

2. Confirm the ESP32 connects to Wi-Fi - you should see `[WiFi] Connected`.
3. Enter a 4-digit booking access code on the keypad followed by `#`.
4. The ESP32 calls `GET /api/iot/verify?code=XXXX&court=1` - if the code matches a confirmed active booking, the servo unlocks the gate for a few seconds then relocks.
5. Trigger the PIR sensor - ESP32 reports occupancy to `POST /api/iot/occupancy`, which updates the admin dashboard.
6. Admin can also manually lock/unlock the gate from the web dashboard - the ESP32 polls `GET /api/iot/gate/status` every 2 seconds.

RUNNING THE FULL SYSTEM

1. Open XAMPP Control Panel → Start MySQL.
2. In terminal, start the backend:

```
``bash
cd server
npm start
``
```
3. In a separate terminal (new tab), start the frontend:

```
``bash
npm run dev
``
```
4. Power on the ESP32 device.
5. Open application in a browser: <http://localhost:5173>

WORKFLOW

1. Register or log in as a user
2. Book a court - select date, time, duration, and court number

3. Upload payment proof (QR or cash) - admin approves to activate the access code
4. At the court, enter the 4-digit access code on the keypad followed by `#`
5. Servo unlocks the gate, PIR sensor reports court occupancy to the admin dashboard

TROUBLESHOOTING

1. `npm` not recognised
 - Verify Node.js is installed:

```
``bash

node -v

npm -v

``
```
 - If not found, download and install Node.js from <https://nodejs.org> and reopen the terminal.
2. Database connection error (`Access denied` or `ECONNREFUSED`)
 - Ensure MySQL is running in XAMPP
 - Verify `DB\_HOST`, `DB\_USER`, `DB\_PASSWORD`, and `DB\_NAME` in `server/.env` match your XAMPP MySQL settings
 - Confirm `database-setup.sql` was imported successfully in phpMyAdmin
 - Default XAMPP MySQL: user `root`, no password - leave `DB\_PASSWORD=` blank if no password is set
3. Frontend cannot reach backend (`Network Error` or blank data)
 - Confirm the backend is running on port 5000
 - Confirm `VITE\_API\_URL` is not set, or is set to `http://localhost:5000`
 - Check that `FRONTEND\_URL=http://localhost:5173` is set in `server/.env`
4. ESP32 Wi-Fi not connecting

- Double-check `WIFI\_SSID` and `WIFI\_PASS` in the firmware
 - Ensure the ESP32 is on the same network as your PC running the backend
 - Use `ipconfig` (Windows) to get your PC's local IP and update `API\_BASE\_URL` in the firmware
5. Keypad code not accepted
- Ensure the booking payment has been approved by admin (access code is only active after approval)
 - Confirm the booking date and time matches the current time
 - Check that `IOT\_DEVICE\_KEY` in the firmware matches `pickleball-iot-prototype-key` (default)
6. Servo / PIR / Keypad not responding
- Verify wiring against the pin table in **Hardware Pin Section**
 - Check that the ESP32 is powered via USB or an external 5V supply
 - Open Serial Monitor (115200 baud) to read debug output from the ESP32

CONCLUSION

After completing all installation steps, the Pickleball Booking System Integration with IoT is fully operational. The system connects a React-based web booking platform with an ESP32-controlled gate, allowing courts to be accessed using booking-generated 4-digit codes verified in real-time against the backend database.